Student Academic Performance and Engagement Data Warehouse: Analyzing At-Risk Factors and Course Outcomes

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ABSTRACT

Early identification of academically at-risk students is critical for implementing timely educational interventions and reducing dropout rates. This case study leverages a multi-source relational dataset comprising student demographics, daily attendance logs, homework submission histories, academic performance records, and teacher-parent communication logs. By merging these disparate data sources on student identifiers, we apply machine learning classification techniques (such as Random Forest or Logistic Regression) to predict student underperformance—defined as falling below a specific grade threshold—prior to final examinations. The findings highlight key behavioral predictors of academic success, offering educators a data-driven framework for proactive student support.

INTRODUCTION

University life is a period marked by significant academic challenges, lifestyle adjustments, and psychological transformations. During this transitional phase, students often face multiple stressors such as academic pressure, financial burdens, sleep disturbances, and changes in dietary habits. These factors can significantly influence their mental well-being and academic performance.

In recent years, there has been an alarming increase in mental health concerns among students, particularly suicidal ideation, often stemming from a complex mix of personal, environmental, and academic variables. This case study seeks to analyze the relationship between various socio-demographic, lifestyle, academic, and psychological variables—including age, study satisfaction, sleep duration, CGPA, work pressure, and family history of mental illness—on the presence of suicidal thoughts.

This study aims to investigate how a combination of academic pressures, lifestyle habits, and psychological factors relate to suicidal thoughts among university students. By identifying the most influential variables, the goal is to provide insights that can support the development of targeted mental health interventions and improve student well-being.

OBJECTIVES

(1) Determine which behavioral factors (e.g., missed classes vs. unsubmitted homework) have the strongest statistical correlation with poor exam scores. (2) Analyze whether proactive teacher-parent communications (teacher_parent_communication.csv) successfully correlate with improved homework and attendance compliance. (3) Provide institutional stakeholders with a threshold-based alert system to flag struggling students before it's too late to intervene.